

Towards a FAIR Framework for Earthquake catalogs in Aotearoa New Zealand

A Community White Paper v0.1

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Executive Summary

Earthquake catalogs are essential for seismology, underpinning research, monitoring, and risk management. In Aotearoa New Zealand, catalogs are developed for diverse purposes, but suffer from uneven access, inconsistent standards, and limited interoperability. Two national workshops in 2023 and 2024 brought together researchers, agencies, and industry to co-design a practical framework guided by the FAIR principles (Findability, Accessibility, Interoperability, and Reusability) and, in the Aotearoa New Zealand context, by the CARE Principles for Indigenous Data Governance (Collective benefit, Authority to control, Responsibility, and Ethics). This white paper outlines the current catalog landscape, community-identified challenges, and a proposed framework with implementation recommendations. A prototype portal has been developed to exercise the framework specifications against representative datasets, demonstrating end-to-end workflows for catalog ingestion, quality assessment, merge and conflation, analytics, and programmatic access. Our aim is to establish a national capability that improves transparency, comparability, and reusability of earthquake data, enabling reproducible science, accelerating research, and strengthening hazard and risk decision-making.

Key Recommendations

1. Adopt a shared catalog metadata profile and controlled parameter lexicon aligned with QuakeML and FDSN conventions, distinguishing required from recommended fields.
2. Require persistent identifiers (DOIs), explicit versioning, and machine-readable changelogs so that any cited catalog version can be retrieved exactly as used.
3. Establish a national “catalog of catalogs” portal and API — complementing, not replacing, GeoNet — with cross-agency event-identifier mapping and transparent merge/conflation.
4. Apply a transparent, configurable data-quality grading scheme and report uncertainties universally, including qualitative confidence classes for historical events.
5. Stand up formal governance (a Steering Committee and a Data Stewardship Team) that operationalises both FAIR and CARE principles, underpinned by sustained multi-year funding.

Revision History

Revision	Date	Author(s)	Description
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1 Introduction

Earthquake catalogs are the cornerstone of seismology, supporting fundamental research, operational monitoring, and risk management. In Aotearoa New Zealand, many catalogs exist, but they are fragmented, inconsistently documented, and not always interoperable. These limitations hinder both research and policy. Ensuring that earthquake catalog data are *Findable, Accessible, Interoperable, and Reusable* is critical to maximise their value in science and practice. Our goal is a nationally coordinated framework and portal that makes catalog data transparent, comparable, and citable.

New Zealand's national operational catalog curated by GeoNet provides broad coverage and public access, including web and programmatic services (GeoNet, 2025a,c,d). Internationally, initiatives such as the European Plate Observing System (EPOS) Seismology service demonstrate the benefits of standardised FAIR data services and coordinated infrastructure (EPOS, 2025b,a). Importantly, the framework proposed here is not intended to replace or duplicate GeoNet's operational mandate; rather, it adds an aggregation and curation layer — a “catalog of catalogs” — that makes GeoNet's national catalog interoperable with the many regional, temporary, historical, and derived catalogs that currently sit outside any common system. This paper synthesises outcomes from two national workshops and intersession working group activity to propose a FAIR framework tailored to the New Zealand context.

2 Current Catalog Landscape

New Zealand's earthquake data ecosystem comprises a set of catalogs that serve different purposes. Understanding their strengths and gaps is a prerequisite to a FAIR framework.

2.1 National Operational Catalog

GeoNet maintains the national operational catalog and associated services, exposing event information to the public and researchers through search tools and standard interfaces (GeoNet, 2025a,c). Strengths include consistent coverage and accessibility throughout the country. Gaps include evolving metadata conventions across time, mixed-magnitude types, and limited exposure of some detailed processing fields to end users.

2.2 Regional and Volcano-specific Catalogs

Research catalogs for specific regions or processes add high-resolution detail. For example, a consistent and high-precision catalog for the Taupō Volcanic Zone demonstrates enhanced detection and relocation accuracy in a volcanically active area

95 (Illsley-Kemp and Mestel, 2025). These catalogs often use dense local networks or specialised methods. Their findability and interoperability can be limited if formats, identifiers, or metadata differ from national standards.

2.3 Temporary Deployments and Experiment Catalogs

100 Following significant earthquakes or for targeted studies, temporary arrays and ocean bottom deployments generate short-term catalogs with lower magnitude thresholds. These datasets are valuable, but are not routinely discoverable or merged with the national catalog without additional curation and metadata alignment.

2.4 Historical and Derived Catalogs

105 Historical catalogs compiled from written and oral records and derived catalogs for hazard modelling or research snapshots complement the instrument era. Without formal versioning, identifiers, and processing documentation, these sets can diverge from operational catalogs and be difficult to reuse or compare (Gerstenberger et al., 2024).

Summary of Strengths and Gaps

- 110
- **Strengths:** depth precision and completeness in local studies; consistency and access in national datasets.
 - **Gaps:** variable completeness through time and space; inconsistent processing; uncertain magnitude and depth scales; missing metadata; opaque methods; uneven access to legacy and temporary data.

3 Earthquake Catalogs: New Zealand and International Practices

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Earthquake catalogs record key details such as time, location, depth, and magnitude, and are essential for seismic hazard analysis, research, and engineering design (Woessner et al., 2010, 2016). New Zealand's catalog, maintained through GeoNet by Earth Sciences New Zealand (ESNZ; formerly GNS Science), has evolved since the 1930s, combining instrumental data and historical records (GeoNet, 2025a,d; GNS Science, 2020). Similar catalogs exist globally (e.g., the United States Geological Survey, the Swiss Seismological Service, and the European-Mediterranean Seismological Centre). This section compares New Zealand's catalog with international counterparts, identifies core attributes of robust catalogs, and outlines steps toward a harmonised global framework for earthquake science and engineering.

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3.1 New Zealand (GeoNet)

GeoNet maintains New Zealand's earthquake catalog as part of the National Earthquake Information Database, serving as the central repository for NZ seismic events (GNS Science, 2020). The catalog contains origin time, location, magnitude, and seismic phase data (GeoNet, 2025a), with each event assigned a unique Public ID (GeoNet, 2025a). Coverage spans instrumentally recorded earthquakes (1930s–present) and historical events from oral/written records (GNS Science, 2020).

Data access includes Quake Search (CSV, GeoJSON, KML, GML) (GeoNet, 2025a) and services compliant with the International Federation of Digital Seismograph Networks (FDSN) and Web Feature Service (WFS) standards, delivering QuakeML with uncertainties, phase arrivals, and multiple magnitude determinations (GeoNet, 2025c,b). The catalog covers NZ and offshore regions, with magnitude of completeness of approximately M_L 1–2 in well-monitored regions (higher elsewhere and through time); location quality and completeness vary markedly with station coverage across the country, a heterogeneity quantified for the GeoNet catalog by Warren-Smith et al. (2025). The catalog features automatic and reviewed solutions (GeoNet, 2025b) and event-type classification (GeoNet, 2025c). A 2022 revision moved the catalog toward more consistent magnitude reporting nationwide (GNS Science, 2020).

3.2 United States (USGS ComCat)

The United States Geological Survey (USGS) operates the federated Advanced National Seismic System (ANSS) Comprehensive Earthquake Catalog (ComCat), covering global earthquakes ($M_{4.5+}$ complete globally; $M_{1-2.5}$ in the US) (USGS, 2025). It is accessed via the FDSN Event API (U.S. Geological Survey, 2025) with QuakeML default output and optional CSV/GeoJSON/KML, plus real-time GeoJSON feeds (U.S. Geological Survey, 2025). Catalog entries include standard parameters plus quality metrics (stations used, azimuthal gap, RMS residuals) and magnitude uncertainty (USGS, 2025).

ComCat integrates 100+ networks (California, Alaska, Hawaii, ISC-GEM, others), storing multiple solutions per event with one flagged as preferred (USGS, 2025). Additional products include ShakeMap and crowdsourced intensity data (USGS, 2025).

3.3 Switzerland (SED)

The Swiss Seismological Service (SED) maintains a high-quality catalog dating to 1879 (Diehl et al., 2025), with historical events to 250 AD. A dense network achieves M_c 1.0–1.5 completeness. SED introduced updated magnitude scales (M_{Lhc} , 2020), employs 3D velocity models and relative relocation, and provides formal uncertainties via QuakeML since 2013 (Diehl et al., 2025).

Data are accessible through FDSN web services. Notable cross-border collaboration integrates 60+ stations from France, Germany, Italy, and Austria for improved border

165 coverage (Diehl et al., 2025). The ECOS project unified historical records into a
homogenised database (Fäh et al., 2011).

3.4 European/International

170 The **European-Mediterranean Seismological Centre (EMSC)** operates the Euro-
pean Seismic Portal, aggregating data from multiple networks via FDSN-compliant
services and providing multiple solutions and phase data (European-Mediterranean
Seismological Centre, 2025a,b).

175 The **International Seismological Centre (ISC)** archives global bulletin data from 130+
agencies, accepting multiple formats (Nordic, GSE, ISF, QuakeML) (International Seis-
mological Centre, 2025a) with output in ISF or QuakeML (International Seismological
Centre, 2025b).

180 Other agencies (the Japan Meteorological Agency, Geoscience Australia, and the
European Integrated Data Archive) have adopted FDSN services, enabling uniform
access. Major catalogs (USGS, GeoNet, SED, EMSC, ISC) now share common
standards for open, interoperable data access (EPOS, 2025b; ETH Zürich Seismology
/ QuakeML Working Group, 2015).

3.5 Gap Analysis: New Zealand vs International Best Practice

185 Table 2 provides a comprehensive comparison of New Zealand's earthquake cat-
alog capabilities against international benchmarks, specifically the USGS ComCat
(United States) and Swiss SED/EPOS systems. The comparison evaluates ten critical
dimensions: core event parameters, metadata and provenance documentation, ver-
sioning and review processes, data access mechanisms, historical data integration,
uncertainty quantification, multi-catalog federation, special-purpose products, stan-
dards compliance, and governance structures. While New Zealand's GeoNet catalog
demonstrates strong technical capabilities, particularly in API provision and QuakeML
190 support, several gaps emerge relative to international practice. Key priorities include:
(1) universal uncertainty reporting across all catalog entries, (2) formalised version
control with Digital Object Identifiers (DOIs) and changelogs, (3) development of an
event-identifier mapping service to link disparate catalogs, (4) stronger integration
of historical and pre-instrumental data, and (5) establishment of formal governance
195 structures with dedicated funding. Addressing these gaps would bring New Zealand's
catalog practices into line with international FAIR data leaders.

200 International earthquake catalogs demonstrate varying scales and capabilities. GeoNet
(NZ) provides regional coverage (M_L 1–2 completeness in well-monitored regions,
1930s+; with spatially variable location quality, Warren-Smith et al., 2025) with unique
Public IDs and QuakeML support (GeoNet, 2025a; GNS Science, 2020). USGS Com-
Cat operates globally ($M_{4.5+}$ complete worldwide, $M_{1-2.5}$ in US, 1900+), federating
100+ networks with ShakeMap products (USGS, 2025). The Swiss SED achieves
exceptional resolution (M_c 1.0–1.5, 1879+ with historical to 250 AD) through 3D models
and cross-border collaboration (Diehl et al., 2025). EMSC rapidly aggregates European

²⁰⁵ data with EventID mapping (European-Mediterranean Seismological Centre, 2025b), while ISC archives global data from 130+ agencies as a long-term reference (1900+) (International Seismological Centre, 2025a).

Table 2: Comparison of NZ and International Earthquake Catalog Capabilities

Feature	New Zealand	USGS ComCat	Swiss SED / EPOS	Gaps & Priorities
Core parameters	Origin time, location, depth, magnitude, phase picks, station counts	Hypocenter, magnitudes, phase data, moment tensors, macroseismic data (USGS, 2025)	Historical + instrumental events, uniform M_w , uncertainties, intensities (Diehl et al., 2025)	Lacks full uncertainty propagation for all events
Metadata & provenance	Some document processing pipelines; incomplete or inconsistent across catalogs	Maintains standards, data dictionaries, method documentation (USGS, 2025)	Adheres to FDSN, StationXML, QuakeML standards (Diehl et al., 2025)	Need consistent, centralised metadata policies
Versioning & review	Rapid + review workflows; some reprocessing	Versioned updates, multiple origins per event, revisions over time (USGS, 2025)	Unified access to evolving datasets, harmonisation	Formalise version control (DOIs, changelogs)
Access & APIs	Quake Search, FDSN, QuakeML, CSV/GeoJSON export	FDSN API, GeoJSON/CSV feeds, search interfaces (U.S. Geological Survey, 2025)	Integrated access to waveforms, parametric data via standards (EPOS, 2025b)	Enhance federated discovery, cross-catalog queries

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Table 2 – continued

Feature	New Zealand	USGS ComCat	Swiss SED / EPOS	Gaps & Priorities
Historical data	Some pre-instrumental events with greater uncertainty	Includes large historic events, variable completeness	Centuries of data, macroseismic fields, uncertainties (Diehl et al., 2025)	Strengthen integration, metadata, uncertainty modelling
Uncertainty & quality	Many provide estimates (location/magnitude error); not universal	Quality metrics: RMS, station count, azimuthal gap (USGS, 2025)	Emphasises uncertainty, solution quality, EPOS standards (Diehl et al., 2025)	Universal uncertainty reporting needed
Multi-catalog linking	Mostly separate; limited cross-catalog mapping	Aggregates many networks, links multiple origins per event (USGS, 2025)	Harmonised catalogs across Europe; cross-border exchange (EPOS, 2025b)	Build Event ID mapping service, federated portal
Special products	Focal mechanisms, moment tensors, slow slip as standalone datasets	Hosts moment tensors, additional products tied to events (USGS, 2025)	Integrates fault, slip, seismological products	Better integrate special-purpose catalogs

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Table 2 – *continued*

Feature	New Zealand	USGS ComCat	Swiss SED / EPOS	Gaps & Priorities
Standards compliance	Supports FDSN/QuakeML; not all catalogs fully adhere	FDSN and QuakeML in APIs and exports (U.S. Geological Survey, 2025)	Full QuakeML, StationXML, FDSN compliance (Diehl et al., 2025)	Ensure full compliance across all catalogs
Governance & sustainability	ESNZ/GeoNet or individual researchers; less formal structure	Formal governance, funding, institutional roles (USGS/ANSS/NEIC) (USGS, 2025)	Robust institutional backing, cross-national coordination	Establish formal governance, roles, funding

4 Workshop Process and Stakeholder Engagement

4.1 Community Workshops

The development of the framework was community-driven. Two pre-conference workshops at the Geoscience Society of New Zealand annual meetings provided the primary engagement: a 2023 workshop framed needs and objectives, and a 2024 workshop reviewed progress and refined priorities (Geoscience Society of New Zealand, 2023, 2024). Participants spanned university researchers, GeoNet and Earth Sciences New Zealand staff, government agencies, and engineering and insurance users. Discussion was structured around eight thematic tables — covering minimum catalog attributes, metadata and documentation, access and versioning, governance, and community priorities — whose outcomes are summarised in Appendix A. Between workshops, a working group met regularly to draft standards, test integration pathways, and design a prototype portal.

4.2 Prototype Portal: Capabilities and Demonstration

To ground community feedback in a working system, the working group developed a functional prototype portal implementing the core workflows proposed in this framework. The prototype serves as a reference implementation that demonstrates the feasibility of the framework specifications and surfaces practical design issues early; it was used at the 2024 workshop to demonstrate ingestion, merge, export, and analytics workflows with representative New Zealand datasets. We describe it here as a proof of concept rather than a validated production service: the capabilities listed in this subsection are implemented in the prototype, whereas the national framework described in Section 5 is a proposal that, except where explicitly noted, is not yet implemented.

Data ingestion and format support. The prototype accepts earthquake catalog data in multiple standard formats: CSV and tab-delimited text files with automatic delimiter and date-format detection, JSON and GeoJSON in several common structural variants, and QuakeML 1.2. A field-mapping interface allows contributors to align source column names to the canonical schema, with alias-based auto-detection to reduce manual effort. Streaming parsers handle large files without excessive memory consumption. Direct integration with GeoNet's FDSN Event Web Service supports incremental import with automatic duplicate detection.

Search, filtering, and visualisation. Users can search, filter, and visualise catalogs through an interactive portal. Spatial queries using bounding boxes, drawn polygons, and radius circles are supported alongside time-range, magnitude-range, and depth filtering. An interactive map provides event clustering, uncertainty-ellipse overlays, focal mechanism visualisation (beach ball diagrams), and station coverage display. Tabular views support pagination and virtual rendering for large event sets.

Catalog merge and conflation. A catalog merge engine allows users to combine multiple catalogs with configurable strategies: priority-weighted (nominated agency

preference), uncertainty-weighted average, quality-based (retaining the higher-quality solution), or complete-event selection (retaining the solution with the most populated fields). The engine detects conflicts based on spatial separation, time offset, magnitude difference, depth range, and network inconsistency. Conflict logs are generated for quality control review and the resulting merged event retains lineage links to its source events. A preview mode allows users to inspect merge results before committing, including a summary of conflicts and quality-control flags.

Analytical tools. Implemented capabilities include Gutenberg-Richter b -value estimation, completeness magnitude (M_c) detection, energy release analysis, cluster detection, and focal mechanism visualisation. Multi-catalog comparison tools support magnitude-frequency distribution (b -value comparison) and uncertainty visualisation across catalogs.

Multi-format export. Catalogs and event subsets are exportable as CSV, GeoJSON, KML, and QuakeML 1.2 from both the portal and the API. Version-specific exports ensure that a cited dataset version can be retrieved exactly as it was at time of citation.

Community feedback from the 2024 workshop informed refinements to the metadata schema, merge configuration options, and quality grading approach described in the remainder of this paper.

5 Proposed FAIR Framework

The framework addresses the end-to-end lifecycle of catalog data. Unless a capability is explicitly identified as implemented in the prototype (Section 4.2), the components below describe a proposed national specification rather than a deployed service. The components are outlined below, and their relationship is summarised in Figure 1.

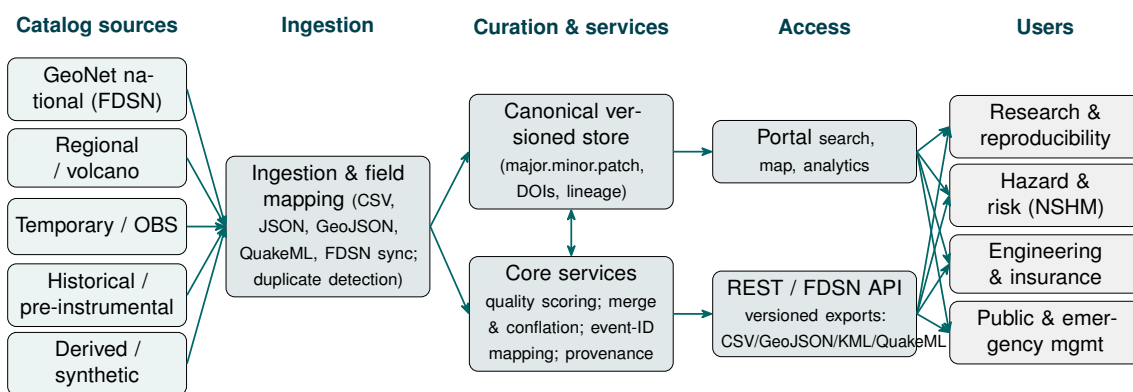


Figure 1: End-to-end architecture of the proposed framework. Heterogeneous catalog sources are ingested and mapped to a canonical, versioned store; curation services (quality scoring, merge and conflation, event-identifier mapping, and provenance) feed a portal and a programmatic API that serve research, hazard, engineering, and public users. Capabilities identified as implemented in Section 4.2 are exercised end-to-end by the prototype.

270 5.1 Metadata Standards for Catalogs

A core metadata profile aligned with QuakeML and FDSN conventions captures identifiers, origin parameters with uncertainties, magnitude types with errors, detection and quality metrics, methodology descriptors, event classification, and catalog-level metadata (ETH Zürich Seismology / QuakeML Working Group, 2015; GeoNet, 2025c; 275 U.S. Geological Survey, 2025). Required versus recommended fields are distinguished to balance utility and effort.

At the catalog level, required metadata include: a unique persistent identifier, catalog name and short description, contributing institution(s), temporal coverage, geographic coverage, magnitude range, event count, processing methodology summary, license, 280 and contact information. Recommended metadata add: velocity model reference, location algorithm, phase picking method, magnitude scale definitions, quality flag descriptions, and links to associated publications.

At the event level, required fields are origin time, latitude, longitude, depth, and at least one magnitude value with its scale type. Recommended fields include origin 285 time uncertainty, horizontal and vertical location uncertainties, magnitude uncertainty, station count, azimuthal gap, phase count, RMS residual, evaluation status (automatic or reviewed), and event type. Table 3 summarises the field hierarchy.

Table 3: Metadata Field Hierarchy for FAIR Catalog Records

Field	Status	Description
<i>Catalog-level</i>		
Persistent identifier	Required	DOI or equivalent resolvable identifier
Name, description	Required	Human-readable title and summary
Institution, contact	Required	Responsible party and contact details
Temporal coverage	Required	Start and end date of events
Geographic coverage	Required	Bounding box or region descriptor
Magnitude range	Required	Minimum and maximum magnitude
Event count	Required	Total number of events
License	Required	Machine-readable license identifier (e.g. CC-BY 4.0)
Processing summary	Required	High-level description of methods
Velocity model	Recommended	Reference to velocity model used
Location algorithm	Recommended	Software and version (e.g. NonLinLoc 7.0)
Magnitude calibration	Recommended	Scale definitions and calibration reference
Publication links	Recommended	DOIs of associated papers
<i>Event-level</i>		
Origin time	Required	UTC time (ISO 8601)
Latitude, longitude	Required	WGS84 decimal degrees
Depth	Required	Kilometres below sea level
Magnitude value & type	Required	Value and scale code (e.g. M_w , M_L)
Origin time uncertainty	Recommended	Seconds (1σ)
Horizontal uncertainty	Recommended	Kilometres (1σ)
Vertical uncertainty	Recommended	Kilometres (1σ)
Magnitude uncertainty	Recommended	Standard deviation
Station count	Recommended	Number of stations used
Azimuthal gap	Recommended	Degrees
Phase count	Recommended	Number of phase picks used
RMS residual	Recommended	Seconds
Evaluation status	Recommended	“automatic” or “reviewed”
Event type	Recommended	QuakeML event type vocabulary

5.2 Shared Parameter Lexicon

A controlled vocabulary standardises magnitude types, event classifications, phase names and quality codes, parameter units and formats, and dataset naming. Using QuakeML event type semantics and community practices improves interpretability and filtering across catalogs (ETH Zürich Seismology / QuakeML Working Group, 2015; Mulder, 2016).

5.3 Programmatic Access and Data Formats

5.3.1 API Design

Unified APIs provide machine-readable access across catalogs, returning CSV, GeoJSON, and QuakeML as appropriate. Bulk downloads of versioned datasets support reproducible research. Services are built on existing GeoNet interfaces and standards (GeoNet, 2025d,c; ETH Zürich Seismology / QuakeML Working Group, 2015).

The API follows REST conventions. Authentication uses token-based credentials for write operations; read operations on public catalogs are unauthenticated. Rate limiting applies to protect service availability. Standard HTTP error codes and a consistent JSON error envelope enable reliable programmatic error handling. Table 4 summarises the principal endpoint categories.

Table 4: Principal API Endpoint Categories

Category	Operations	Notes
Catalogs	List, retrieve, create, update, delete	Metadata including bounds, event count, license
Events	List, spatial query, time-range, magnitude/depth filter	Pagination; supports GeoJSON, CSV, QuakeML output
Search	Cross-catalog full-text and parameter search	Returns ranked results across all accessible catalogs
Import	Upload file, sync from FDSN source, view history	Supports CSV, JSON, GeoJSON, QuakeML 1.2
Merge	Preview, execute, retrieve conflict log	Dry-run mode returns statistics without committing
Export	Per-catalog or per-query download	CSV, GeoJSON, KML, QuakeML; version-specific
Versions	List versions, retrieve specific version	Version-specific event snapshots remain available indefinitely

Output format is negotiated via the `Accept` header or a `format` query parameter, defaulting to GeoJSON for spatial queries and CSV for tabular exports. QuakeML output is generated from the canonical internal event model, preserving uncertainty fields, evaluation status, and magnitude type.

5.3.2 Cross-Catalog Event ID Mapping

A known challenge in multi-agency seismology is that the same earthquake carries different public identifiers across agencies. For example, a New Zealand earthquake may appear as a GeoNet Public ID, a USGS event code, and an ISC event number. Without a mapping service, researchers aggregating data from multiple sources must resolve duplicates manually.

The framework includes an event identity mapping service that maintains a table of cross-agency equivalences. Events are matched using configurable spatial, temporal, and magnitude thresholds (illustratively, separation < 20 km, time offset < 30 s, magnitude difference < 0.5). These values are deliberately permissive to absorb the differing location procedures used across agencies; the trade-off is that, in dense aftershock sequences, wide windows can group genuinely distinct events. Thresholds are therefore configurable per source, and ambiguous matches are flagged for expert review rather than merged automatically. The API exposes equivalence queries, allowing downstream tools to retrieve all known agency identifiers for a given event, deduplicate federated results, and trace the origin of merged solutions to their contributing source records.

Event ID mapping records include the matching agency, the matched identifier, a confidence score, and the thresholds applied. This service aligns with the approach used by EMSC (European-Mediterranean Seismological Centre, 2025b) and extends it to the New Zealand context, facilitating interoperability with USGS, ISC, and regional networks.

5.4 Portal and Infrastructure for Data Discovery

5.4.1 Discovery and Search

A central portal offers unified search and mapping, rich metadata views, download and API endpoints, and documentation. The architecture federates operational feeds and curated research datasets. Integration with international systems enhances global interoperability (EPOS, 2025b,a).

The portal search interface supports free-text search, spatial filtering (bounding box, polygon, radius), time-range and magnitude-range filtering, depth filtering, and event-type filtering. Saved filter configurations allow users to reproduce a specific query for monitoring or repeated analysis. Cross-catalog search returns a merged, deduplicated result list ranked by relevance.

5.4.2 Catalog Merge and Conflation

Catalog conflation — combining events from multiple independent catalogs into a unified view — is a central operation in multi-agency seismology. The framework specifies the following merge strategies, all four of which are implemented in the prototype (Section 4.2); users select a strategy based on their scientific objective:

- **Priority-weighted:** a nominated agency's solution is preferred; solutions from other agencies are recorded as alternatives.
- **Uncertainty-weighted average:** location, time, and magnitude are computed as an inverse-variance-weighted mean across contributing solutions.
- **Quality-based:** the solution with the highest quality score (see Section 5.9) is retained.
- **Complete-event:** the solution with the greatest number of populated recommended fields is retained.

Conflicts are detected and logged when contributing solutions disagree beyond configurable thresholds: spatial separation, temporal offset, magnitude difference, depth range, and network mismatch. Each merged event retains explicit lineage to its contributing source events and the merge configuration used, enabling full audit and reproducibility. A dry-run preview mode returns conflict statistics and quality flags without committing a new catalog version.

As a worked example, Table 5 shows a single earthquake reported by two agencies and the result of an uncertainty-weighted merge. Each merged parameter is an inverse-variance-weighted combination of the contributing solutions, the combined uncertainty is reduced accordingly, and the merged record retains lineage links to both source events.

Table 5: Worked example: uncertainty-weighted merge of one event reported by two agencies (illustrative values).

Field	Source (GeoNet)	A	Source B (USGS)	Merged (uncertainty-weighted)
Event identifier	2016p858000		us1000778i	merged-0001 (lineage to A, B)
Origin time (s past 11:02:00 UTC)	56.3 ± 0.5		56.7 ± 0.8	56.4 ± 0.4
Latitude (°)	−42.737		−42.740	−42.738
Longitude (°)	173.054		173.100	173.061
Depth (km)	15.1 ± 2.0		15.0 ± 4.0	15.1 ± 1.8
Magnitude (M_w)	7.8 ± 0.1		7.8 ± 0.1	7.8 ± 0.07
Horizontal (km)	unc. 1.5		3.0	1.3

5.4.3 Analytics

The portal provides in-browser analytical tools to support quality assessment and comparative analysis. Capabilities already implemented in the prototype (Section 4.2) include Gutenberg-Richter b -value estimation with confidence intervals (Aki, 1965), completeness magnitude (M_c) detection (Wiemer and Wyss, 2000), energy-release

summaries, spatial cluster detection, focal-mechanism visualisation, and magnitude-frequency distribution comparison across multiple catalogs. These tools enable researchers to identify potential biases or artefacts before downloading data for further analysis.

5.5 Licensing and Access Conditions

5.5.1 License Requirements

To support FAIR reusability, each dataset and metadata record should carry an explicit, machine-readable license and any access constraints (Wilkinson et al., 2016; GO FAIR Initiative, 2025). For government-funded baseline products, a default open license aligned with New Zealand Government Open Access and Licensing guidance is recommended, with clearly documented exceptions for embargoed, sensitive, or culturally restricted data (New Zealand Government, 2022). License metadata should be exposed through portal views and APIs to avoid ambiguity for downstream reuse.

5.5.2 Role-Based Access Control

Access to catalog data and portal operations is governed by a role-based model. Four roles are defined:

- **Administrator:** full access including user management, system configuration, and deletion of catalog records.
- **Editor:** can create and modify catalogs, import data, and initiate merge operations, but cannot manage users or alter system configuration.
- **Viewer:** can search, view, filter, and export catalogs to which they have been granted access, but cannot modify data.
- **Guest** (unauthenticated): can discover and download public catalogs; embargoed or restricted catalogs are not visible.

Role assignments are managed by Administrators and are recorded in an audit log. A role-upgrade request workflow allows registered users to request elevated access from an Administrator, with notifications and approval tracking. These four roles describe *technical* access to the portal and are distinct from the *governance* roles (contributor, reviewer, maintainer) used in the catalog contribution and stewardship process (Section 6.1.3); a single individual may hold both a system role and a governance role.

5.5.3 Data-Level Access Control

Access control applies at the catalog level. Each catalog carries one of three access designations: **public** (accessible to all without authentication), **registered** (accessible

to authenticated Viewers and above), or **restricted** (accessible only to named individuals or groups, as set by the Administrator). This mechanism supports time-limited embargoes, for example allowing a research group to work with a catalog privately until a companion paper is published.

Field-level masking may be applied where a catalog contains sensitive location information, such as event locations near culturally significant sites. In such cases, location coordinates may be spatially generalised or withheld for Guest and Viewer access while full precision is retained for authorised users, consistent with the CARE framework described in Section 5.11.

5.5.4 Audit and Compliance

All write operations (catalog creation, import, merge, deletion) are recorded with the authenticated user identity, timestamp, and operation details. Read access to restricted catalogs is similarly logged. These records support institutional accountability and can be provided to data owners on request. Where usage agreements are attached to a catalog (for example, a requirement to cite a companion publication), the agreement is presented at download time and acceptance is recorded.

5.6 FAIR Compliance Matrix and Indicators

A practical FAIR compliance matrix maps framework requirements to the FAIR sub-principles (F1–F4, A1–A2, I1–I3, R1.1–R1.3) (Wilkinson et al., 2016; GO FAIR Initiative, 2025). Table 6 gives an illustrative mapping from each sub-principle to the framework mechanism that addresses it and a measurable indicator that can be reported annually. Reporting these indicators annually would provide transparent evidence of progress and guide prioritisation.

Table 6: Illustrative FAIR Compliance Matrix: Sub-Principle, Framework Mechanism, and Indicator

Sub-pr.	Framework mechanism	Indicator
F1	Persistent identifiers (DOIs) for catalogs; stable event identifiers	% of catalogs with a resolvable DOI; % of events with a stable ID
F2	Rich metadata profile (catalog- and event-level)	Mean metadata-completeness score across catalogs
F3	Metadata records embed the data identifier	% of metadata records linking to their data identifier
F4	Indexing in the searchable portal and discovery API	% of catalogs indexed and discoverable
A1	Retrieval via FDSN/REST over HTTPS	API uptime; % of catalogs retrievable by identifier
A1.1/A1.2	Open, standard protocols; token auth only for write/restricted data	% of read endpoints requiring no authentication
A2	Metadata and version records retained after data withdrawal	% of withdrawn datasets with surviving metadata
I1	QuakeML / FDSN standard representations	% of catalogs available as valid QuakeML
I2	Controlled parameter lexicon (magnitude, event, phase types)	% of fields using controlled-vocabulary terms
I3	Qualified cross-references (event-ID mapping, lineage links)	% of merged events with recorded source lineage
R1.1	Machine-readable license on every record	% of catalogs with a machine-readable license
R1.2	PROV-based provenance (entities, activities, agents)	Mean provenance-completeness score
R1.3	Conformance to FDSN/QuakeML community standards	% of catalogs passing standards-compliance checks

5.7 Versioning and Persistent Identifiers

5.7.1 Versioning Policy

Each catalog release carries a version and timestamp, with change logs. Significant data sets receive DOIs to enable citation. Event identity is preserved through stable identifiers and cross-references, with aliasing where needed. All versions remain accessible for reproducibility.

A formal versioning policy distinguishes patch-level revisions (metadata corrections or other non-substantive edits) from major and minor revisions, which change the citable data content — by altering event parameters or processing methods (major), or by adding new events or optional fields (minor) — and therefore warrant a new version

DOI (DataCite, 2025). Version links should be recorded in metadata (for example, relations equivalent to “IsNewVersionOf” and “IsPreviousVersionOf”) to maintain a clear lineage of record.

The framework adopts a three-level versioning scheme: MAJOR.MINOR.PATCH. Major version increments indicate changes that alter event parameters for existing events (e.g., full reprocessing with a new velocity model or magnitude calibration). Minor version increments indicate additions of new events or new optional fields without changing existing event solutions. Patch increments indicate metadata corrections, documentation updates, or bug fixes that do not affect event parameters. Table 7 gives illustrative examples.

Table 7: Versioning Scheme: Examples of Version Increment Triggers

Version	Trigger	Example change
MAJOR	Event parameters changed	Reprocessing with new 3D velocity model; recalibration of magnitude scale; adoption of new location algorithm affecting systematic offsets
MINOR	New events or fields added	Ingestion of 500 additional aftershock events; addition of focal mechanism fields to existing events
PATCH	Metadata or documentation	Contact email updated; description clarified; deprecated field removed from catalog-level record

5.7.2 DOI Assignment and Changelog

The catalog series as a whole is identified by a single concept DOI, while each major and minor release receives its own version DOI minted through a recognised registration agency; patch releases update the existing version record without minting a new DOI. To avoid an unmanageable proliferation of identifiers where new events are added frequently, minor releases may be batched on a regular cadence (for example, scheduled quarterly snapshots) rather than minted per ingestion. DOI metadata includes catalog name, version, release date, geographic and temporal coverage, responsible institution, license, and relations to predecessor and successor versions.

A machine-readable changelog accompanies each release, recording the type of change (added, updated, fixed), a description, and summary statistics (total event count, new events, modified events). Changelogs are published alongside the catalog record and retrievable via API. Together, versioned DOIs and changelogs allow a researcher to cite the exact data state used in a study and allow reviewers to understand what changed between referenced versions.

5.8 Documentation of Processing, Provenance, and Uncertainty

Processing steps are documented, including detection, picking, location algorithms, velocity models, and magnitude calibration. Uncertainties for time, location, and magnitude are quantified or qualitatively flagged. The provenance, evaluation status, and links to raw waveforms and related products are included.

To strengthen reproducibility, provenance fields should follow a consistent model describing entities (input data and outputs), activities (processing steps), and agents (people, institutions, and software), with software version and execution date captured as minimum requirements (World Wide Web Consortium, 2013). This structure supports transparent audit trails across operational and research catalogs.

5.8.1 Uncertainty Representation

Uncertainty quantification is essential for scientific reusability. The framework specifies a standard representation to ensure consistency across catalogs and comparability in merged products.

Origin location and time uncertainties are reported as standard deviations: `time_uncertainty` (seconds), `horizontal_uncertainty` (kilometres), and `depth_uncertainty` (kilometres). Where horizontal uncertainty is anisotropic, the semi-major and semi-minor axes of the 68% confidence ellipse and its azimuth are reported. These map directly to QuakeML `OriginUncertainty` elements (ETH Zürich Seismology / QuakeML Working Group, 2015).

Magnitude uncertainties are reported as the standard deviation of the magnitude estimate on the stated scale. Because uncertainty is scale-dependent (local magnitude estimates carry different error profiles from moment magnitude estimates), the magnitude type code must always accompany the uncertainty value.

Events without quantified uncertainties are flagged with a boolean `uncertainty_available` field set to false. For historical or pre-instrumental events where formal uncertainty estimation is not possible, a qualitative confidence class may be substituted (e.g. “well-constrained”, “poorly-constrained”, “felt-only”). The data quality scoring described in Section 5.9 penalises records where uncertainty information is absent.

Propagation through merges. When events from multiple catalogs are merged, the combined uncertainty depends on the merge strategy. For uncertainty-weighted averaging, the combined standard deviation is computed from inverse-variance weighting. For priority or quality-based strategies, the selected solution’s uncertainty is retained and the alternatives recorded. The merge log documents which strategy was applied and the resulting uncertainty for each merged event.

5.9 Data Quality Assessment Framework

5.9.1 Quality Scoring Methodology

A standardised quality scoring system enables consistent evaluation of catalog completeness and reliability, supporting informed reuse decisions. Quality is assessed across five dimensions:

1. **Completeness:** fraction of required and recommended fields populated (weighted: required fields carry higher weight than recommended fields).
2. **Accuracy:** internal consistency checks, including depth–magnitude plausibility, non-negative uncertainty values, and coordinates within the catalog’s declared geographic bounds.
3. **Reliability:** proportion of events with reviewed (versus automatic) solutions, station count and azimuthal gap relative to thresholds, and phase count adequacy.
4. **Consistency:** temporal ordering, absence of duplicate events (by time and location), and cross-field agreement (e.g. magnitude type codes matching declared scale definitions).
5. **Traceability:** availability of processing documentation, at least one contact record, and a link to the catalog version record or companion publication.

An overall quality score (0–100) is computed as a weighted sum across dimensions (illustrative weights: 40% completeness, 20% accuracy, 15% reliability, 15% consistency, 10% traceability). These weights, and the grade thresholds in Table 8, are an illustrative starting point rather than an empirically calibrated standard: they should be tuned by the community against representative New Zealand catalogs, and the scoring scheme is designed so that both weights and thresholds are configurable. The score maps to a letter grade as shown in Table 8.

Table 8: Quality Grade Thresholds and Recommended Use (illustrative; subject to community calibration)

Grade	Score	Interpretation and recommended use
A+	95–100	Excellent. Publication-ready; suitable for hazard modelling and all quantitative analysis.
A	85–94	Good. Suitable for most research and engineering applications.
B+	75–84	Acceptable. Minor gaps; document limitations in methods sections.
B	65–74	Fair. Notable gaps; consider supplementing with additional metadata.
C+	55–64	Poor. Significant gaps; expert review recommended before quantitative use.
C	45–54	Very poor. Limited utility for quantitative analysis.
D	35–44	Minimal utility. Use only for discovery or illustrative purposes.
F	<35	Not recommended for quantitative analysis.

Quality grades are computed and displayed per catalog in the portal, and per event where event-level quality flags are populated. A quality report exportable as JSON or CSV provides a full breakdown of scores, flag descriptions, and actionable recommendations for catalog contributors.

5.9.2 Specific Quality Checks

Representative automated checks include: events ordered chronologically; no future-dated events; depth within a physically plausible range for the region (e.g. -5 to 1000 km, the negative bound allowing for events located above the sea-level datum); magnitude within a plausible range for the declared region; uncertainty values non-negative and within parameter-specific sanity bounds (for example, horizontal uncertainty not exceeding the catalog's declared spatial extent, and magnitude uncertainty below a configurable ceiling such as 1.0 magnitude unit); no coordinate pair repeated for distinct events at an identical time (duplicate detection); station count ≥ 5 for the event to be considered well-constrained; and azimuthal gap $< 180^\circ$ for reliable depth estimates (USGS, 2025). Results are summarised as passed, warning, or failed for each check category.

5.9.3 Anomaly Detection

Statistical outlier detection flags events whose parameters fall outside expected distributions for the catalog as a whole. Events with unusually high magnitudes relative to surrounding events in time and space, or with depths inconsistent with the regional tectonic setting, are flagged for expert review. Flagged events are not removed automatically; the system annotates them so that users can apply appropriate caution.

5.10 Special Catalog Types and Integration

Different catalog types require adapted handling to preserve scientific value while meeting FAIR standards.

5.10.1 Regional Research Catalogs

Regional catalogs produced with specialised methods (e.g. 3D velocity models, relative relocation, custom magnitude calibrations) are ingested as distinct catalogs with their own version records and DOIs. Processing methods are documented in the catalog-level metadata, including references to the velocity model and location algorithm used. Regional catalogs are not automatically merged into the national operational catalog because different processing assumptions may introduce systematic biases; instead, the portal provides cross-reference links between co-temporal events in regional and national catalogs, with the spatial and magnitude offset reported so that researchers can assess consistency.

5.10.2 Temporary Deployment Catalogs

Catalogs from aftershock deployments or targeted experiments are ingested with an explicit time window and a designation of “research” or “monitoring” type. Station geometry metadata (array bounds or individual station codes) is stored as catalog-level attributes. Where waveform data are archived, links to the relevant FDSN waveform service are included. Temporary deployment catalogs are searchable and downloadable alongside permanent catalogs but are labelled to indicate their limited geographic and temporal scope.

5.10.3 Historical and Pre-Instrumental Catalogs

Historical catalogs present distinct challenges: location precision may be tens of kilometres, magnitude estimates are converted from macroseismic intensity or historical accounts, and formal uncertainty estimation is often not possible. The framework accommodates these catalogs through:

- Qualitative location confidence classes (e.g. “well-constrained”, “approximate”, “felt-only”) in place of or alongside numeric uncertainties.
- Support for macroseismic intensity fields (Modified Mercalli or EMS-98 (Grünthal, 1998)) alongside or in place of instrumental magnitude.
- Narrative description fields for source attribution (e.g. newspaper records, missionary accounts, oral tradition references).
- Separate quality thresholds for historical records, reflecting their inherently lower parameter precision.

Historical catalogs are not merged with instrumental catalogs by default; they are presented as a separate collection with explicit data type labelling. Cross-reference links to instrumentally recorded aftershocks or re-analyses may be added where they exist.

5.10.4 Derived and Synthetic Catalogs

Derived catalogs (filtered subsets, hazard snapshots, or synthetic seismicity forecasts) are ingested with mandatory provenance metadata identifying the source catalog(s), derivation method, parameters, and software. Derived catalog records carry explicit “derivedFrom” relations to their source catalog versions, enabling users to trace a derived product back to the underlying data. Synthetic catalogs used in probabilistic seismic hazard analysis are labelled as “synthetic” to distinguish them from observational records.

5.11 Aotearoa New Zealand Data Governance Context

In Aotearoa New Zealand, the framework should be implemented in a way that is consistent with both FAIR and CARE principles, so that data stewardship recognises Indigenous rights, authority, and benefit alongside openness (Carroll et al., 2020). For catalogs incorporating mātauranga-informed interpretations, iwi-relevant records, or sensitive contextual information, governance processes should document appropriate access conditions, attribution, and engagement pathways.

Where event records relate to locations of cultural significance or incorporate traditional ecological knowledge, the framework’s data-level access control (Section 5.5.3) provides one technical mechanism to restrict or mask sensitive fields. This is, however, only a mechanism: the appropriate access conditions for such records must be determined with, not for, iwi and hapū, who should be engaged as partners and decision-makers from the outset rather than consulted as data subjects after the design is fixed. The Steering Committee’s membership (Section 6.1.1) should include representation from relevant Māori communities and data sovereignty advocates to ensure that CARE principles are operationalised rather than aspirational.

6 Implementation Recommendations

6.1 Governance, Roles, and Responsibilities

6.1.1 Steering Committee

A Steering Committee provides strategic oversight and ensures community ownership of the framework. Membership should include representatives from Earth Sciences

610 New Zealand, at least two universities with active seismology programmes, a gov-
ernment user (e.g. National Emergency Management Agency), an engineering or
insurance sector representative, an Indigenous data sovereignty advocate, and an in-
ternational advisor from a peer organisation (e.g. ISC, EMSC, or GFZ). The Committee
615 meets at minimum quarterly, with defined decision-making processes and staggered
three-year terms to ensure continuity.

The Committee is responsible for approving new data sources and catalogs for ingestion, ratifying versioning decisions that affect the national record, overseeing international partnerships and alignment with EPOS or similar consortia, and commissioning annual reporting against FAIR compliance indicators.

620 **6.1.2 Data Stewardship Team**

A Data Stewardship Team led by Earth Sciences New Zealand handles day-to-day operations. Indicative resourcing for a mature service includes: a Data Stewardship Lead (oversight, Steering Committee liaison), Data Managers (catalog ingestion, meta-
data curation, quality review), a Software Engineer (API maintenance, infrastructure,
625 security), and a User Support Analyst (community queries, documentation, training). Budget should cover staffing, infrastructure (compute, storage, backup), and biennial framework reviews.

6.1.3 Catalog Contribution Process

External researchers and institutions may submit catalogs for inclusion. Submissions
630 require: catalog name and description, responsible institution and contact, time period and geographic coverage, event count and magnitude range, format and metadata documentation, and a declared license. A two-stage review process applies: technical QC (format validity, completeness score, duplicate check) followed by scientific peer review (methods assessment, comparison with existing catalogs). The expected review
635 timeline is four to eight weeks, with written feedback provided regardless of outcome. Approved catalogs are assigned a DOI and version record.

6.1.4 Success Metrics and Annual Reporting

Progress should be measured and reported annually against indicators including:
640 number of catalogs ingested; total event count and geographic coverage; proportion of catalogs with quality grade A or above; API uptime (target: 99.5%); registered user count; number of data downloads and API queries; and number of peer-reviewed publications citing the platform or a versioned catalog DOI. Reporting against FAIR sub-principle indicators (Section 5.6) should also be included. Annual reports should be published openly to maintain transparency and accountability.

645 Illustrative two-year targets: the full GeoNet national catalog (well over one million events) plus 20 or more additional regional, temporary, and historical catalogs accessible through a single interface; 200 or more registered users drawn from the New

Zealand and international research, engineering, and agency communities; 10 or more research outputs citing a versioned catalog DOI; and API availability exceeding 99% (building toward the 99.5% operational target). These figures are indicative planning targets, not commitments, and should be revisited once the service is established.

6.2 Timeline for Development and Adoption

Phase 1: Planning and mobilisation (0–6 months). Formalise governance and resources. Finalise the metadata schema and technical architecture. Establish the Steering Committee and Data Stewardship Team.

Phase 2: Prototyping and initial integration (6–18 months). Build core portal and APIs. Integrate pilot catalogs (GeoNet national catalog plus 5–10 regional and temporary datasets). Refine standards based on contributor feedback. Deliver community training workshops.

Phase 3: Full-scale implementation (18–36 months). Ingest major regional and temporary datasets. Harden and launch the public portal. Activate event ID mapping service. Establish DOI minting workflow.

Phase 4: Maintenance and improvement (36+ months). Operate as a national core service with annual review, FAIR compliance reporting, and planned capability enhancements informed by community feedback.

6.3 Funding and Support Considerations

Secure multi-year funding through research infrastructure programmes and operational budgets. Budget for data curation and development roles, storage and delivery costs, and user support. Indicative resourcing for a mature service is on the order of three to four full-time-equivalent (FTE) staff — a stewardship lead, data managers, a software engineer, and part-time user support (Section 6.1.2) — together with cloud or on-premises infrastructure for storage, compute, and backup, and recurring DOI registration fees. A realistic path begins with a smaller team during Phases 1–2 and scales to the full complement at Phase 3. The figures here are indicative of scale rather than costed estimates; precise budgets should be developed with host institutions. Measure success with adoption metrics, integrated catalog counts, and demonstrable impact on science and hazard modelling.

7 Benefits

Reproducibility. Versioned datasets with persistent identifiers and clear processing history enable verification and reuse.

Science and innovation. Integrated and standardised data reduce barriers and accelerate discovery, aligned with international best practice (EPOS, 2025b).

Risk and hazard. Trusted, timely, and comparable inputs improve hazard and risk assessments and operational products (Gerstenberger et al., 2024).

Collaboration and efficiency. Shared rules, tools, and data reduce duplication and strengthen national capabilities.

8 Conclusion and Call to Action

New Zealand has the data, the expertise, and a clear community mandate to make its earthquake catalogs FAIR; what is missing is a coordinating framework and the governance to sustain it. This white paper has set out that framework, benchmarked it against international practice, and demonstrated its feasibility through a working prototype. The remaining step is collective commitment.

We invite GeoNet and Earth Sciences New Zealand, universities, government agencies, iwi and hapū partners, and the engineering and insurance sectors to:

1. endorse the metadata, lexicon, versioning, and quality standards proposed here as a shared national baseline;
2. commit to establishing the Steering Committee and Data Stewardship Team (Section 6.1.1), with representation that operationalises both FAIR and CARE; and
3. support a multi-year funding case to move from prototype to a sustained national service.

We welcome feedback on this draft and expressions of interest in participating in the governance and catalog-contribution processes described above. Comments may be directed to the corresponding author.

Data and Code Availability

In keeping with the FAIR principles it advocates, the material supporting this white paper is intended to be openly available. The prototype portal source code, the canonical metadata schema, and the controlled parameter lexicon are to be released under an open license at [repository URL to be inserted], and a citable version of this document will be assigned a DOI ([DOI to be inserted]). The earthquake catalogs used as examples are available from their respective providers: the GeoNet national catalog (GeoNet, 2025a), USGS ComCat (USGS, 2025), and the Swiss Seismological Service (Diehl et al., 2025).

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Appendix A: Summary of Workshop Tables 1–8

This appendix summarises the full set of workshop discussions captured across eight interactive tables. The content reflects community perspectives on minimum catalog attributes, metadata requirements, processing concerns, governance needs, and priorities for a national FAIR-aligned framework for earthquake catalogs in Aotearoa New Zealand.

Table 9: Summary of Key Themes from Workshop Tables 1–8

Theme	Key Points
Minimum Catalog Attributes	Phase picks; station metadata; velocity models (1D/3D); depths and magnitudes with uncertainties; completeness estimates; region of interest; unique IDs; focal mechanisms; processing history; detection and processing methods.
Metadata & Documentation Needs	Clear methodology (location algorithm, velocity model, filters); solution provenance; waveform availability; descriptive summaries; parameter definitions; assumptions (point-source vs 2D/3D); embargo details; QC information; required README and DOI; multiple contacts.
Access & Versioning	Raw data accessibility; open formats (CSV, QuakeML, FDSN); programmatic access (API); search by region, time, magnitude; support for multiple versions; backward/forward compatible formats; tracking of method changes; long-term storage of legacy releases.
Governance Needs	Defined roles (contributor, reviewer, maintainer); central catalog portal; consistent naming conventions; periodic updates; upload validation; framework transparency; responsibility for long-term stewardship.
Community Priorities	National catalog portal; common parameter lexicon; national 3D community velocity model; honest uncertainties; digitisation & reprocessing of legacy data; cross-catalog interoperability; clear documentation for older datasets.
Concerns Raised	Inconsistent completeness; variable processing; missing metadata; uncertain magnitude/depth scales; “black-box” ML outputs; compatibility issues across regional/temporary catalogs; handling of unpublished datasets; need for quality flags and user warnings.
Desired Portal Features	Merge multiple catalogs; support subcatalogs; synthetic catalogs; geographic reference tools; method-based filtering; visualisation; minimum-requirements upload form; linkage among event, waveform, and station data.
What Users Liked	Simple form; unified access; clear uncertainty fields; reproducibility focus; increased awareness of catalog availability; prioritisation of essential fields.
Requested Improvements	More metadata fields and filters; better uncertainty representation; link with 2D/3D models; clarify naming/versioning; transparent ML integration; improved handling of old locations; inclusion of focal mechanisms and strong-motion connections.

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